

Response to referee comments:
PRD Submission DM13138,

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October 9, 2024

Author response to the referee report for the paper DM13138, entitled ‘*Adaptive cancellation of mains power interference in continuous gravitational wave searches with a hidden Markov model*’.

We thank the referee for their constructive review. We agree with all the comments and have made the recommended changes as described below. The referee comments are reproduced in italics for convenience. We separate the response into “main” comments (indexed alphabetically) and “additional” comments (indexed numerically), following the convention of the referee.

Referee comments

Main comments

- A. *In section II, it is not sufficiently clear if the model is intended as a realistic representation of the North American power grid, or just a phenomenological model that is "good enough" for the purpose of subtraction. This should be spelled out better. An authoritative reference on the actual grid's design and/or statistical properties, if the authors can find one, would also be highly valuable for the typical target audience of this article (astrophysicists / data analysts) who are unlikely to have such information at hand.*

Author response:

Yes, we agree with the referee that it is important to specify clearly that the model is intended as a phenomenological model that seeks to reproduce the main qualitative features of the power grid (i.e. an approximately constant central frequency with small, slow, periodic modulations; a voltage with a stochastically wandering phase offset; see third paragraph in Section IIB). We have followed the referee's suggestion and updated the discussion in Section II to make this point explicitly.

We also agree that the addition of a reference describing the statistical properties of the grid would be valuable. We have updated the main text to point to two useful papers (Folgueras et al. 2017, Schafer et al. 2018). However, we caution that there is no simple statistical model for frequency fluctuations. The power grid responds to demand by adjusting the frequency. When load increases, the frequency drops, more fuel is supplied to turbine-based generators to meet the increased load, and the frequency rises. The frequency statistics therefore reflect hard-to-predict fluctuations in demand, which stem from random exogenous spatio-temporal factors, such as weather, economic activity, faults, and so on. The challenges of providing a single authoritative statistical model for the properties of the grid are discussed in the cited papers, and we have also included an additional discussion in the main text (Section II).

- B. *In the end, it is not clear enough how big the positive impact of the method really is. Especially, and regardless of the answer to point A, it seems necessary to present at least some level of real-data test of the proposed method, beyond the presented simulated noise case. Since LIGO data is readily available via the GWOSC site, this should be straightforward to perform. A clearer summary of the results in abstract and conclusion would then be appreciated.*

Author response:

Yes we agree that testing the method on real data is a valuable addition. We have added an additional section (Section VI) to the main text where we extend the previous validation tests to real data. We obtain strain data from the LIGO-Livingston strain channel and reference data from the nine independent PEM channels that directly measure mains voltage. We then inject a synthetic GW signal. Before applying ANC the 60 Hz line corrupts the performance of the HMM, which erroneously tracks the noise line. After applying ANC the 60 Hz line is removed and the HMM successfully tracks the underlying GW signal, and the time-averaged root-mean-square error in the frequency estimate is 0.06 Hz.

We also agree with the referee that a clearer summary of the main results quantifying the impact of the method is important to include. We have updated the abstract and the conclusion accordingly. Specifically,

- For representative synthetic data (Table 1), ANC suppresses the 60-Hz interference by approximately 40 dB.
- For representative synthetic data, without ANC, the HMM tracker is unable to track the wandering CW signal. After ANC, the HMM tracker is able to track the wandering CW signal. The time-averaged RMS error in the HMM tracking after ANC is 0.038Hz for low σ_f ($\sigma_f N^{1/2} \Delta t^{1/2} f_{ac}^{-1} = 0.047$; see Section IV B) and 0.47Hz for high σ_f ($\sigma_f N^{1/2} \Delta t^{1/2} f_{ac}^{-1} = 0.15$).
- For real LIGO data (Section VI), ANC suppresses the 60-Hz interference by approximately 20 dB.

- For real data, before applying ANC the 60 Hz line corrupts the performance of the HMM, which erroneously tracks the noise line. After applying ANC the 60 Hz line is removed and the HMM successfully tracks the underlying GW signal, and the time-averaged root-mean-square error in the frequency estimate is 0.06 Hz.
- Tests on synthetic data confirm that the detection probability at a 5% false alarm rate satisfies $p_d(0.05) \approx 0.5\text{--}0.6$ across a range of plausible mains power interference parameters and filter parameters, e.g. $0.5\text{Hz} \leq \Delta f_{ac} \leq 1.5\text{Hz}$

Additional comments

1. *The abstract currently points out that the validation test is on "synthetic signals", which is perfectly fine, but what is more important is the nature of the noise data.*

Author response:

Yes, we agree that the behaviour and statistical properties of the noise are important. We have added a qualifying sentence to the abstract that specifies the properties of the noise.

2. *A brief summary of the results would be valuable at the end of the abstract (as per point B above).*

Author response:

Yes, we agree that the main results should be better summarised in the abstract. We have updated the abstract accordingly as recommended, in line with point B.

3. *A few comments on references the introduction section: The first 3 are very old, describing glitches in initial detectors - consider updating to 2G references like the Davis+ papers already cited later. The standard Advanced Virgo detector reference arxiv:1408.3978 seems to be missing. Please check if [21] (Lee et al) was indeed the intended citation - this seems to be a deep learning waveform model paper, not about artifact identification. At the end of the second paragraph, another important class of line vetoes not cited yet are those based on multi-detector consistency (e.g. doi:10.1088/0031-8949/90/12/125001 and doi:10.1103/PhysRevD.89.064023)*

Author response:

We thank the referee for bringing to our attention references which we had previously overlooked, and the erroneous [21] Lee et al. citation. We have now included all the suggested references, removed reference [21], updated to advanced detectors, and included discussion on the class of vetoes based on multi-detector consistency.

4. *In the paragraph below, the statement that candidates are vetoed blindly without checking the strength of the noise contribution is not true for all standard CW pipelines. More nuanced discussions can be found in some recent LVK results papers.*

Author response:

Yes we agree with the referee that the previous statement was imprecise.

We have updated the paragraph to better describe the vetoing procedures for standard CW pipelines (e.g. *FrequencyHough*, *SkyHough*, see arXiv 2201.00697) and discuss "line robust" algorithms, which combine statistics from different detectors to suppress single detector artifacts (e.g. Keitel et al. 2015, Keitel 2016a,b).

5. *Please try using multiple linestyles for better readability of figures 1,6,8,9. In particular, in Fig. 6 it is difficult to discern that the blue $x(f)$ curve actually shows the signal peak, as it is mostly hidden behind the green $h(f)$.*

Author response:

We have trialled different linestyles in the figures to improve readability, as suggested. However, in practice we find that different linestyles do not improve the figure readability. For example, in Fig 6 changing the linestyle of the green $h(f)$ curve still obscures the blue $x(f)$ signal peak. Modifying other figure settings such as the line width also does not help. We

agree with the referee that there is the potential for confusion with Fig 6. We have updated the figure caption to explicitly state that the two peaks are overlaid and difficult to resolve separately by eye.

6. *At the end of II.A, there should be an astrophysical reference for the Crab's frequency, not just a software paper. (Or both together, since citing software used is indeed commendable.) Since the Crab's frequency evolves relatively quickly, a reference year would also be appropriate when quoting it with decimal digits.*

Author response:

We have updated the reference to the Crab frequency to include both an astrophysical reference and a software reference. We now also quote the Crab frequency for a given reference epoch (48442.5 MJD), as suggested.

7. *Fig. 1: missing definition of the abbreviation "ASD"*

Author response:

The abbreviation ASD is now defined in the caption of Figure 1.

8. *After Eq. (1) the strict separation of frequency and amplitude modulation is misleading, as both parts of the Earth's motion contribute to both modulations.*

Author response:

The text has been updated to remove the misleading separation between the two modulations.

9. *Before Eq. (4), when talking about the coupling "in various complicated ways", would this not naturally be expected to cause stochastic overlaps of components with different time delays τ ? Could the authors comment on whether this is a likely explanation for the observed losses in coherence?*

Author response:

The two primary mechanisms by which mains power couples into the strain channel are the shared power supply of the electronic components, and the 60 Hz magnetic field coupling to magnets which are present in certain optical components of the interferometer.

For both the above coupling mechanisms, the time delay τ is the same. Specifically, τ is the delay due to the spatial propagation of the signal between the PEM position and the GW sensing apparatus.

We have added a sentence to the final paragraph of Section IIB to clarify the above point and to flag that other coupling mechanisms may have different τ values, potentially leading to stochastic overlap of different time-delay components, as the referee suggests.

As is discussed in the text, coherence < 1 occurs due to (i) the non-linear relationship between the strain and PEM channels and; (ii) additional unmodelled components (e.g. noise) that are present in one channel but not the other.

10. *footnote 3: gitlab file URLs can be quite ephemeral. Is there no better (more stable) way to reference this list, such as a GWOSC release with DOI, a LIGO DCC entry, etc?*

Author response:

Yes, we agree that it is desirable to provide a more permanent reference to the PEM channels. However, unfortunately at this point in time there is no DOI /DCC reference that covers all nine PEM channels. It is possible to reference the GWOSC O3 auxiliary channel data release, DOI: 10.7935/3bh6-0k98. Note that this dataset only contains a subset (three) of the (nine) PEM channels. For lack of a better option, we have followed the referee's suggestion and included this DOI as an additional reference.

11. *The statements at the end of II.C and in the Fig. 4 caption contradict each other: are the higher-frequency peaks harmonics of the power mains, or "of different provenance"?*

Author response:

Yes, this is a mistake. The higher-frequency peaks are harmonics. We have updated the Figure 4 caption accordingly. We thank the referee for bringing this mistake to our attention.

12. *In the coherence discussion, it remains unclear what threshold is considered to be still sufficient, and below which the channel would no longer be considered a reliable witness.*

Author response:

Yes, we agree with the referee that the performance of the ANC scheme is directly related to the degree of coherence between the strain and witness channels.

Specifically, the fraction of possible noise cancellation R for an idealised system is related to the coherence by

$$R(f) = \frac{1}{1 - |C_{xr}(f)|}$$

see e.g. DOI: 10.1109/ASPAA.1997.625628 or DOI: 10.3390/s22176591.

The threshold value of $R(f)$ required in order for HMM algorithms to run successfully is an empirical question that depends on the properties of the signal (e.g. how strong is the signal? How much does it stochastically wander?). We have added a discussion on these points to Section IIC.

13. *In Eq. (12), it is not made explicit what argument the “arg min” is taken over. Would it be appropriate to write $e_k(w)$ with an explicit argument?*

Author response:

Yes, we agree with the referee’s suggestion. We have updated Equation (12) to specify explicitly that the argmin is taken over the argument $\mathbf{w}$ (i.e. the tap weights), which influence the residual $e_k(\mathbf{w})$.

14. *In IV.A, can the authors make an argument why neglecting the Earth’s movement is not changing qualitatively the signal-vs-line behaviour? Since CW signals are expected to modulate very slowly intrinsically, typically the shape and extent of their detector-frame frequency spectrum should be dominated by these terms.*

Author response:

The referee is correct to point out that the Earth’s motion would influence the received GW signal. However, continuous wave searches based on the F-statistic matched filter, such as those analysed in this paper, correct for the Earth’s movement automatically by including the Earth-related terms in the phase model tracked by the F-statistic. For the HMM used in this paper, the detection statistic at each step of the algorithm is a function of the calculated F-statistic values only; the Earth’s motion does not enter anywhere else. Hence neglecting the Earth’s motion does not have any influence on the efficacy of the ANC algorithm, which is the main focus of this paper (see also point 15).

15. *Also in the same discussion, there is a reference to the F-statistic. This is taking a maximum likelihood over amplitude parameters, not over those related to the Earth’s movement. Indeed F-statistic searches typically also include a “demodulation”, which is probably what the authors had in mind here. But the way it is currently phrased will confuse a non-expert reader into thinking it maximizes over frequency evolution parameters.*

Author response:

Yes, the referee is correct to point out the potential for confusion through the current phrasing. We have rephrased these sentences to clarify that demodulation is readily included in standard continuous wave search pipelines.

16. *In IV.B, the definition of signal-to-noise ratio does not match the standard in the CW literature, which usually includes the observation time. The duration is also important for the frequency resolution. Is it the 800s visible in Fig. 7? Please clarify more explicitly.*

Author response:

Yes, we agree with the referee that traditionally the definition of SNR in the CW literature does include the value of T_{obs} . In this paper we fix $T_{\text{obs}} = 800\text{s}$ throughout and vary only the ratio h_0/σ_n . Accordingly, we previously felt that for this paper it was sufficient to neglect

the scaling with T_{obs} in the definition of the SNR. However, we agree with the referee that in order to make better contact with the CW literature it is more appropriate to include T_{obs} in the SNR definition. We have updated the relevant definitions in the manuscript accordingly. Regarding the question on whether the $T_{\text{obs}} = 800\text{s}$ is visible in Figure 7, indeed the vertical axis covers the 800s observation period. We have updated the main text to clarify this point more explicitly, as suggested.

17. *The case (ii) "not plotted for brevity" would seem relevant to show - since now fgw lies within the affected frequency range, it sounds like it could be qualitatively different, and more impressive to demonstrate that the method can still deal with it?*

Author response:

We appreciate the referee's recommendation and did our best to implement it. However in practice plotting case (ii) produces a plot which is almost identical to Figure 6, but with the injected signal (solid green curve) having a broader spectrum (see Figure 1 in this document). Whether we use case (i) or case (ii) does not affect the ability of the ANC to suppress the 60 Hz noise line. Actually the success of the ANC/HMM method for case (ii) is demonstrated nicely by Figure 7, where the HMM is able to successfully track the GW signal (dashed white curve and solid green curve overlap in right-panel).

18. *At the end of IV.B, have the authors verified that the Earth Doppler effect is negligible compared to these frequency errors?*

Author response:

Yes, via previous work (Suvorova et al. 2016, 2017). We have now added a sentence making this point (with the associated references) in Section IV B. As noted in point 14, the Earth's Doppler effect only enters through the well-tested F-statistic, which corrects for it. It is not referenced directly by the ANC algorithm (or even the HMM, actually).

19. *In figures 8/9, it would be interesting to also compare with the performance without ANC - is that similar to, or even worse than the "random guess"?*

Author response:

When we run the HMM algorithm to try to recover the GW signal that overlaps with the 60 Hz instrumental line, the algorithm erroneously tracks the 60 Hz line. Accordingly, the HMM consistently claims an erroneous detection and the performance is worse than a random guess. That is, the "performance without ANC" $p_d - p_{\text{fa}}$ curve reduces to the single point $p_d = 0$ and $p_{\text{fa}} = 1$.

20. *Why is the "zero interference ROC curve" shown only in Fig. 9, not 8? Is it understood why the best performing ANC results are still quite far away from this, if before the subtraction looked visually very impressive?*

Author response:

Yes, this is a mistake. We agree that the zero interference curve should be plotted for both figures. We have updated Figure 8 accordingly.

Under some circumstances, the ANC does not perform as well as for zero interference, simply because the noise is not being removed perfectly. The performance of the ANC filter can be improved in general by tuning the relevant parameters. For example in the bottom panel of Figure 9, the $\lambda = 1$ ROC curve is very similar to the zero-interference curve, for reasons that are discussed in the text. We have added additional discussion on this point to the main text in Section V.A.

21. *Near the end of V.B, latency issues are discussed. Are these really relevant for CW searches, which are usually run on archival data?*

Author response:

The referee is correct to point out that latency is not a problem when dealing with offline searches. We have removed the discussion on latency from the manuscript and now focus exclusively on the computational cost of the method.